

HANDOFF REPORT: DEEP LOGICAL REASONING

Calibrated Optimal Stopping, Confidence Metrics & Dynamic Strategy Routing

Branch: `exp/calibrated-stopping-logic` | Commit: `07de8c1` | Base: `main` | Date: 06/10/2026

1. Executive Research Overview

This branch contains the mathematical foundation and experimental core for the NCKH research study on **budget-aware dynamic reasoning**. The core hypothesis tests whether dynamically routing and escalating reasoning strategies (CoT → Self-Consistency → Candidate Selection) only when confidence is low achieves a superior Pareto trade-off between task accuracy and inference tokens compared with static one-shot strategies.

Mathematical Objective: $\max_{\pi} E [\text{Acc}(\pi(q)) - \lambda \cdot \text{Cost}(\pi(q))]$

2. Critical Challenges Resolved

A. Confidence Recalibration (Platt Scaling & Isotonic Regression)

Development audits revealed that all incorrect answers had raw heuristic confidence ≥ 0.80 . Without recalibration, threshold fitting collapsed with $\tau \approx 1.0$ (always escalating). We added **PlattCalibrator** (logistic regression) and **IsotonicCalibrator** (monotonic regression), along with **brier_score** and **expected_calibration_error** (ECE) metrics in `optimal_stopping.py`.

B. Bootstrap Confidence Intervals on Stopping Thresholds (τ)

Single point estimates for τ lacked statistical uncertainty bounds. We implemented **bootstrap_thresholds()** with B resamples with replacement, producing point estimates, standard deviations, and empirical 95% confidence intervals `[ci_lo, ci_hi]`.

C. Unified Extraction & Sample-Weighted Consensus

Integrated **extract_item_answer()** to automatically dispatch schema-aware typed rules (number, text, expression, decimal comma) and extended **majority_vote(weights=...)** to support confidence-weighted self-consistency consensus.

D. 6-Dimensional Entry Complexity Features

Expanded **complexity_features(version=2)** with ordering cues ('before', 'after', 'rank'), conditional logic ('if', 'unless', 'given that'), and syntactic nesting depth, while preserving exact 3D legacy compatibility when `version=1`.

E. Configurable Initial Confidence Reward Shaping

In ReasoningEnv, step rewards previously ignored confidence gains on step 1 (where `had_answer` was False). Added **reward_initial_confidence** to award $\beta \cdot \text{conf}$ on the initial step in RL training.

3. Modified Artifacts & Responsibilities

File	Key Additions / Roles
<code>optimal_stopping.py</code>	Platt/Isotonic calibrators, ECE, Brier score, bootstrap_thresholds, transform_conf
<code>reasoning_strategies.py</code>	extract_item_answer, weighted majority_vote, safe math.isclose checks
<code>reasoning_env.py</code>	6D expanded complexity features, reward_initial_confidence parameter in gym Env
<code>pipeline.py</code>	check_answer item schema integration, negation false-positive guard
<code>test_optimal_stopping.py</code>	Tests for Brier, ECE, Platt/Isotonic error reduction, bootstrap CIs
<code>test_reasoning_env.py</code>	Tests for 6D complexity features, initial confidence step reward shaping
<code>test_reasoning_strategies.py</code>	Tests for extract_item_answer (number/comma/text), weighted majority vote

4. Verification & Testing Evidence

• **Core Research Regression Suite:** `python -m pytest tests/test_research_study.py test_optimal_stopping.py test_entry_predictor.py -q` → **29/29 PASSED** (5.31s)

- **Extended Headless Research Tests: 125/125 PASSED** (15.26s)
- **Synthetic Smoke & Replay:** `python -m experiments.research_study --backend smoke` and exact replay matched byte-for-byte with zero deviation.

5. Recommended Next Steps

1. **Recalibrated Threshold Sweep:** Run tuning sweeps with `--recalibrate platt` on candidate dataset splits.
2. **Manuscript Output:** Tabulate Pareto curves and bootstrap intervals ($\tau \pm CI_{95}$) for the NCKH paper.
3. **Stage 0 Gate:** Ensure train, calibration, and test splits remain strictly disjoint before real Apple Silicon model execution.